Designing an Academic Coaching Platform for Students with Executive Dysfunction

A Comparative Analysis of Web-Based Tools and a Potential New Solution

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IST 477 Capstone

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Abstract

Executive dysfunction affects 30-50% of college students with ADHD and related conditions (Cleveland Clinic, n.d.), yet current productivity tools fail to comprehensively address their unique cognitive and emotional barriers. This capstone project investigates which features of existing productivity tools help or hinder students with executive dysfunction, identifies which of the six executive function components (Activation, Focus, Effort, Emotion, Memory, Action) are most neglected by current solutions, and proposes improvements through the development of a web-based academic coaching platform.

Through comparative analysis of nine widely used productivity tools (Google Calendar, Notion, Todoist, Trello, phone reminders, Pomodoro apps, ChatGPT, habit trackers, and paper planners), this research reveals that while tools excel at Memory support (average 3.9/5), they systematically fail at Activation (2.9/5) and Emotion (2.9/5) (See Appendix B). The analysis identifies several critical design failures: complexity burdens that create high activation energy, fragmentation problems requiring multiple apps, and emotional neglect through shame-inducing interfaces.

To address these gaps, this project developed a prototype academic coaching website (https://lhorowi.github.io/capstone/) that integrates low-activation-energy design principles, emotional safety features, and comprehensive support across all six executive function components. Key innovations include one-click task capture, AI-assisted assignment breakdown, non-judgmental "crisis mode" interfaces, and integrated focus tools—all designed specifically for neurodivergent students.

This work contributes a replicable framework for evaluating productivity tools through an executive function lens, demonstrates how to design intentionally for students with executive dysfunction rather than merely accommodate it, and provides a foundation for accessible, evidence-based educational technology that bridges the gap between clinical academic coaching ($100-200/hour) and inadequate productivity apps.

Chapter 1: Introduction

1.1 Personal Connection and Project Idea

It's a Thursday night, 11:47 PM. A student has an essay due Friday night at 11:59 PM that they have known about for three weeks. Their laptop is open to a blank Google Doc. They have made a cup of tea, organized their desk, created a new Notion template, updated their Todoist, and checked their calendar twice. Somehow, it is now 12:30 AM and they still haven't written a single word. This is not procrastination. This is paralysis.

This is what executive dysfunction can feel like.

I've been that college student who opens their assignment with genuine intentions, only to spend hours deciding on the "perfect" organizational system for that assignment rather than engaging with the work itself. Meanwhile, the actual assignment that prompted all this organization sits untouched. When one part of the assignment becomes too overwhelming, I shift my attention to another, often something easier or more immediately rewarding. For a while, this feels productive. I'm adding tasks, checking them off, and regaining a sense of control. Then life happens, and I miss a few days, overdue tasks start piling up, and those notification badges become constant reminders of my perceived failures. This then triggers a "shame spiral" where guilt and self-criticism make it increasingly difficult to re-engage. I abandon my original strategies, but without a clear way to break down that assignment into manageable steps, I end up back where I started, just staring at a blank screen, paralyzed.

This cycle isn't unique to me. Research shows that students with ADHD experience "app fatigue"—frequently trying and abandoning multiple productivity tools in search of one that

works (Ali & Thu, 2025), each promising to be "the one" that finally works. But they don't work. It is not because students with these executive function issues are not trying hard enough, but because they're not designed for how our brains function.

My personal breaking point came during my junior year, when I nearly failed a class. It was not because I couldn't do the work, but it was because for some reason I could not get myself to turn it in. I had the knowledge, the time, even the motivation, and most of the work done. What I lacked were tools that worked with my cognitive processes instead of against them. That's where the central idea of this capstone came from: asking myself the question "What if we could understand exactly why productivity tools fail students with executive dysfunction, and use that understanding to design something more effective?"

1.2 What is Executive Dysfunction/Executive Function Disorder

Executive dysfunction is not about intelligence, laziness, or lack of motivation. It's a neurological difference in how the brain's executive functions operate, which are the cognitive processes that allow one to plan, organize, initiate tasks, sustain attention, regulate emotions, and monitor their own behavior (Barkley, 2012; Brown, 2013).

Executive functions are controlled by the prefrontal cortex, part of the brain responsible for higher-order thinking and self-regulation. In individuals with executive dysfunction, these processes do not function in the same way that they do for neurotypical individuals. For example, in someone with ADHD, the brain's dopamine regulation differs from regular patterns.

This affects motivation, reward processing, and the ability to sustain attention on tasks that aren't immediately engaging (Cleveland Clinic, n.d.; Barkley, 2012).

Additionally, the prefrontal cortex is one of the last parts of the brain to fully mature, with development continuing through age 25 (Diamond, 2013). As a result, neurotypical college students are still developing these skills even under typical circumstances, while students with executive dysfunction are facing additional neurological barriers.

While executive dysfunction is most associated with ADHD (affecting 30-90% of individuals with ADHD), it is not exclusive to ADHD (Cleveland Clinic, n.d.). It also affects people with autism spectrum disorder, anxiety, depression, learning disabilities, traumatic brain injuries, and other neurotypical individuals under extreme stress (Cleveland Clinic, n.d.; Diamond, 2013). The scope of students who could benefit from better executive function support is much larger than just those with ADHD diagnoses.

Compounding this issue, approximately 80% of students report never being taught how to study, manage time, or plan ahead (Branching Minds, n.d.; Understood.org, n.d.). These skills are necessary for college success and directly rely on executive function abilities. College students are expected to have these skills, yet rarely receive instruction in them, and they are later penalized when they struggle.

To develop a deeper understanding of executive dysfunction, one must first understand the six components of executive function: Activation, Focus, Effort, Emotion, Memory, and Action (Brown, 2013; CHADD, n.d.). Each of these components presents distinct challenges and requires different support strategies.

1. **Activation** relates to getting started. This can include organization, prioritization, and task initiation. Difficulties with activation may appear as staring at a blank page, unable to start; knowing exactly what you need to do but feeling frozen; spending hours "preparing" to work without working; or avoiding an assignment that has been assigned for weeks.
2. **Focus** relates to the part of oneself associated with concentrating on relevant information or filtering out distractions. Challenges with focus may include rereading the same paragraph five times without absorbing anything from it; intending to work but realizing twenty minutes later that the time has been spent scrolling social media or feeling attention constantly pulling toward anything more immediately stimulating.
3. **Effort** relates to the parts of oneself associated with maintaining effort and processing speed. Difficulties with effort may appear when a student feels mentally exhausted after twenty minutes of work, is unable to push through when tasks require sustained thinking, experiences a sense that their brain just simply stopped cooperating, or struggles to process information at the pace it's presented.
4. **Emotion** relates to the ability to manage one's feelings and frustrations. This includes regulating emotional responses, handling setbacks without emotional derailment, or even maintaining motivation despite challenges. Challenges with emotional regulation may present when a student receives a bad grade and it leads to thoughts of dropping out, when overwhelming shame about being behind that prevents a student from catching up, when small setbacks feel catastrophic, or when anxiety becomes so intense that it creates paralysis.

5. **Memory** relates to holding and retrieving information. This addresses both using working memory to hold information as well as accessing recall or prospective memory. Difficulties in this area can be seen when one walks into a room and forgets why they are there, writes down an assignment and then forgets the paper exists, struggles to hold multi-step instructions in their head, or when one constantly loses personal items.
6. **Action** relates to monitoring and self-regulating behavior. This includes impulse control, flexibility to adapt strategies when they're not working, self-monitoring and correction, or breaking tasks into steps. For students with executive dysfunction, challenges with action may present as submitting work without proofreading, getting stuck on one approach even when it is not working, not knowing how to break an assignment down into concrete steps, or a difficulty learning from mistakes.

1.3 Current Approaches to Supporting Executive Dysfunction

At the start of this capstone, the original intention was to simply find the best existing resources for students with executive dysfunction. Many support strategies have been developed, including professional academic coaches, campus disability support programs, and individual self-help strategies.

Academic coaching has emerged as one of the most effective forms of executive function support. Academic coaches work with students to help them meet both academic and personal goals, such as helping students build the skills to be able to break large assignments down into manageable pieces, creating additional accountability and developing strategies to manage both academic demands and overall wellbeing. They also support students in learning how to

advocate for their needs and accommodations. Campbell and Mogashana (2024) found that academic coaching significantly supports student success through these structured and guided approaches.

While academic coaching has proven to be an effective resource for goal clarification and skill development, it is also a very limited resource. Academic coaches are expensive, typically charging between $100-200 for an hour-long session. They can only take on a limited number of students and have limited hours. If a student were to have a crisis at 11 PM on a Sunday night, an academic coach would not be expected to be available to help.

Syracuse University's OnTrack program illustrates these constraints (Syracuse University CDR, n.d.). OnTrack provides academic coaching primarily to first-year students with an option for them to continue through their second year. Students meet twice a week with their academic coach at a cost of $3,200 for the semester, or approximately $110 per session. As Campbell and Mogashana (2024) note, academic coaching is not scalable. Even well-established programs like OnTrack at Syracuse University face these scalability challenges, since each coach can only take a limited number of students.

Many students with executive dysfunction receive accommodations through campus disability services. Students can access services like extended time on exams, reduced course loads, early power-point access, and flexibility with deadlines. However, accessing accommodations requires diagnosis, documentation, and self-advocacy. It also requires that students understand what accommodations are available, what they are eligible for, and what they might need.

Other students develop their own informal strategies for their academic success. To help accountability, they may work with study groups or use body doubling techniques where they work in the presence of another person, not for help or collaboration, but simply for shared presence and accountability. For focus, they may change their environment, such as going to the library instead of studying in a dorm room. They may even develop routines or habits when it comes to their study patterns. While these strategies can work, they rely heavily on the student's ability to initiate, maintain and adapt them independently.

These strategies are not as reliable as an academic coach since they require significant self-knowledge, self-control, and trial-and-error. Maintaining this consistency is significantly harder for students with executive dysfunction. Even worse, when a student with executive dysfunction is going into "crisis mode" and truly needs the support, the self-regulation required to maintain these informal strategies is precisely what executive dysfunction impairs, leaving students without support at their most vulnerable moments.

1.4 Identified Gap and Project Rationale

Despite the growing number of digital tools marketed toward students with executive dysfunction, none offer a comprehensive system that supports all six components of executive function while also addressing the emotional and contextual realities of academic life. Existing tools primarily optimize for task tracking and memory support, assuming users can independently initiate action, manage emotions related to productivity, and consistently engage with the system. For students with executive dysfunction, these assumptions are often invalid, leading to tool abandonment, increased shame, and heightened academic stress.

This project responds directly to these gaps by proposing a unified, student-centered academic support website designed specifically for executive dysfunction rather than adapted from professional productivity systems. Unlike existing tools, this system prioritizes low activation energy, emotional safety, and academic specificity from the outset. By integrating task breakdown, planning assistance, reminder support, and crisis-tolerant design into a single platform, the project aims to reduce cognitive overload, minimize shame, and support students throughout the full academic workflow.

Chapter 2: Comparative Analysis

2.1 Productivity Tools Market and User Patterns

The productivity tools market has exploded, with thousands of apps currently available. Some of the most popular productivity tools include Google Calendar, Notion, Todoist, Trello, Phone Reminders, Pomodoro Apps, AI Tools, Habit Trackers, and Paper Planners. Students with ADHD report higher abandonment rates than neurotypical users. After a student with executive dysfunction finds a new tool that they believe will fit their needs, they usually follow a cycle of going through a honeymoon period, slowly finding flaws and other reasons to stop using the product, and then finding their next support tool. This cycle is referred to as "app fatigue" which applies to both frustration with the quantity of apps available as well as the quality of them (Ali & Thu, 2025). App fatigue reflects the frustration that students with executive dysfunction can have about having all of their academic tools spread across different applications and websites. Students often use several different productivity tools at the same time to maintain their lives. This only further shows the fundamental mismatch between how tools are designed and how executive dysfunction manifests.

2.2 Tool-by-Tool Analysis

The analysis of the nine different productivity tools for this Capstone looks at how they perform in supporting everyday executive function skills. Using custom rating criteria (see Appendix B, Table B.1), each tool was scored on a five-point scale across the six areas of executive function. Memory, Action, Focus, Effort, Activation, and Emotion, plus three additional factors: Usability, Effectiveness, and Accessibility. This approach made it easier to see not just which tools work

well overall but also where they shine or fall short in helping people manage their thinking, organization, and motivation.

When the results were combined (see Table C.10, Appendix C), some clear patterns emerged. The tools were strongest in supporting memory and Action, meaning they're generally good at helping people remember information and follow-through on tasks. Usability also scored well, showing that most tools are relatively easy to navigate. However, Activation and Emotion rated the lowest, suggesting that while the tools can help with organization and follow-up, they're less effective at encouraging users to get started or stay motivated. These findings highlight where current tools meet users' needs and where there is still room for improvement.

2.2.1 Google Calendar

Overall, Google Calendar earned an average rating of 4 out of 5 (see Table C.1, Appendix C). It is excellent for time management but weak when it comes to actual task execution. It is an excellent external memory aid and time-awareness tool, but it provides little support for planning, task initiation, or sustained execution.

- Activation (4/5): Very quick to add events with little setup, but a blank calendar can feel overwhelming and requires users to make decisions about what to schedule.
- Focus (3/5): Helpful visual tools like color-coding and multiple views but overlapping events and notifications can create clutter and distractions.
- Effort (4/5): Low maintenance once set up with repeat events and syncing, but only effective with consistent input and attention to reminders.

- Emotion (3/5): Neutral and non-judgmental with customization options, yet overscheduling and past-due events can feel overwhelming and anxiety-inducing.
- Memory (5/5): Strong reminder system with notifications, integrations, and location-based alerts, though too many can cause fatigue.
- Action (4/5): Clearly shows when to work through time-blocking and conflict alerts but offers no guidance on how to complete tasks.

2.2.2 Notion

Overall, Notion earned an average rating of 2.3 out of 5 (see Table C.2, Appendix C). It is extremely flexible and powerful, but highly overwhelming for users with executive dysfunction. It functions well as a centralized information and planning space, but provides little support for activation, focus, or sustained task execution. The high level of customization shifts cognitive labor onto the user, making it difficult to start, maintain, and follow through on work.

- Activation (2/5): Notion offers templates and flexibility, but too many options create decision paralysis and require significant setup before the tool is usable.
- Focus (2/5): Customizable pages can reduce distractions, but nested pages, sidebar clutter, and multiple views make it easy to get lost and distracted.
- Effort (2/5): Once perfectly set up it can be efficient, but maintaining databases, tags, and structure requires constant effort and ongoing reorganization.
- Emotion (2/5): Aesthetic customization can feel motivating, but perfectionism, comparison to others' setups, and abandoned pages often lead to guilt and avoidance.
- Memory (3/5): Stores large amounts of information in one place with strong search tools but lacks robust reminders and relies on the user to remember to check it.

- Action (2/5): Supports detailed planning and task tracking but does not prompt action or guide users on what to do next, often turning planning into procrastination.

2.2.3 Todoist

Overall, Todoist earned an average rating of 4.1 out of 5 (see Table C.3, Appendix C). It is one of the strongest tools available for executive function support, particularly for task capture, memory, and action initiation, though emotional challenges remain. It functions well as an external task manager that reduces cognitive load and decision fatigue, but it still relies on the user to break down complex tasks and manage emotional responses to overdue items.

- Activation (4/5): Fast task capture through natural language input, shortcuts, and voice entry, but deciding how to organize tasks into projects or labels can still cause users some hesitation.
- Focus (4/5): Clear “Today” section and filtered views highlight what needs attention now, but large task lists and multiple high-priority items can still feel overwhelming.
- Effort (4/5): Low ongoing maintenance with recurring tasks, templates, and quick completion, but users must regularly use to prevent task buildup.
- Emotion (3/5): Checking off tasks and productivity feedback can be motivating, but visible overdue tasks, streaks, and scores can trigger guilt, pressure, or shame.
- Memory (5/5): Excellent reminder system with time-based, location-based, and recurring alerts that significantly reduces mental load, though excessive notifications can cause fatigue.
- Action (4/5): Strong at surfacing clear next actions and prioritization but does not automatically break complex tasks into steps or provide guidance on how to begin.

2.2.4 Trello

Overall, Trello earned an average rating of 3.3 out of 5 (see Table C.4, Appendix C). It is highly effective for visual thinkers and project-based work, but less supportive for daily task execution. It functions well as a visual workflow and project management tool, but provides limited support for consistent activation, reminders, and sustained daily follow-through for users with executive dysfunction.

- Activation (3/5): Quick and satisfying card creation with templates and drag-and-drop interaction, but it requires upfront decisions about structure that can cause decision paralysis.
- Focus (3/5): Visual boards clearly show workflow stages, but card pileups, long lists, and multiple boards can create visual clutter and overwhelm.
- Effort (3/5): Reorganizing cards feels easy and rewarding, but maintaining boards, moving cards, and managing clutter requires ongoing effort.
- Emotion (4/5): Visually tracking progress and moving cards to “Done” is motivating and less judgmental than numeric task lists, though abandoned boards can still induce guilt.
- Memory (3/5): Cards store detailed information and due dates, but reminders are easy to miss and require users to actively check boards.
- Action (3/5): Clearly supports project workflows and task breakdowns but does not prompt specific actions and is better suited for projects than daily execution.

2.2.5 Phone Reminders

Overall, , Phone Reminders earned an average rating of 4.2 out of 5 (see Table C.5, Appendix C). It is deceptively powerful for executive function support due to its extreme simplicity, instant

accessibility, and strong reminder system. It functions exceptionally well as an external memory aid and activation tool, though it provides limited support for complex planning or multi-step task execution.

- Activation (5/5): Instant capture through voice commands, lock-screen access, and zero setup makes it the gold standard for task initiation.
- Focus (4/5): Minimal interface and simple lists reduce distractions, though limited organization options prevent big-picture planning.
- Effort (5/5): Requires virtually no effort to use or maintain, with automatic syncing and one-gesture completion.
- Emotion (4/5): Non-judgmental and pressure-free design avoids shame or performance anxiety, though it lacks encouragement or motivational feedback.
- Memory (5/5): Excellent reminder system with persistent, time- and location-based notifications that provide strong external memory support.
- Action (4/5): Prompts clear, immediate actions with low friction, but cannot break down complex or abstract tasks.

2.2.6 Focus To-Do

Overall, Focus To-Do earned an average rating of 4.3 out of 5 (see Table C.6, Appendix C). It is exceptionally effective at supporting sustained attention and task execution once work has begun, but it provides little support for planning or deciding what to work on. It functions best as a focus enforcement and execution tool, rather than a planning or organizational system for users with executive dysfunction.

- Activation (4/5): One-tap start, and minimal setup lower the barrier to beginning work, though users must still decide what task to work on and how long to set the timer.
- Focus (5/5): Forced focus periods, visual timers, and app or website blocking create strong external structure that supports sustained attention and reduces distractions.
- Effort (4/5): Simple start/stop interface and automatic breaks keep cognitive load low during use, but each session still requires intentional initiation.
- Emotion (4/5): Gamification and visual rewards make work feel more approachable and motivating, though timer pressure and interrupted sessions can create guilt or anxiety.
- Memory (3/5): Tracks focus history and patterns, but offers no reminders or prompts to start working, relying on the user's prospective memory.
- Action (5/5): Highly effective at forcing immediate action and sustaining effort during work sessions but does not support task planning or breakdown.

2.2.7 ChatGPT

Overall, ChatGPT earned an average rating of 4.0 out of 5 (see Table C.7, Appendix C). It is exceptionally effective for planning, problem-solving, and breaking down complex tasks, but provides little support for execution tracking or prospective memory. It functions best as a cognitive support and planning assistant that reduces mental load, rather than a tool for reminders, task tracking, or sustained follow-through for users with executive dysfunction.

- Activation (5/5): Conversational, non-judgmental interface removes intimidation and allows users to start immediately without needing to know the "right" way to ask for help.

- Focus (3/5): One-on-one interaction reduces choice paralysis and supports clarification, but conversations can wander, and long responses may reduce sustained attention.
- Effort (5/5): Offloads significant cognitive work by generating plans, outlines, and step-by-step guidance with no setup or maintenance required.
- Emotion (5/5): Highly supportive, patient, and encouraging, reducing shame and anxiety often associated with executive dysfunction, though it may feel impersonal to some users.
- Memory (2/5): Maintains short-term conversational context, but offers no reminders, notifications, or long-term task tracking.
- Action (4/5): Excellent at creating actionable plans and troubleshooting obstacles but remains passive and requires external tools to track progress and prompt action.

2.2.8 Habitica

Overall, Habitica earned an average rating of 3.8 out of 5 (see Table C.8, Appendix C). It is highly effective for building consistency and reinforcing daily routines, but its reliance on streaks can create emotional pressure and guilt when habits are broken. It functions well as a habit-formation and routine-support tool but offers limited flexibility for variable tasks or changing academic demands common among students with executive dysfunction.

- Activation (3/5): Daily prompts and notifications encourage engagement, but requiring consistent check-ins and upfront habit definition can create friction and decision fatigue.
- Focus (4/5): Clear daily habit lists and one-at-a-time completion reduce overwhelm, though the system is less effective for non-habitual or variable tasks.
- Effort (4/5): One-tap checkoffs and simple interactions keep effort low, but the tool depends on consistent daily logging to remain effective.

- Emotion (5/5 / 1/5): Gamification, rewards, and streaks are highly motivating when maintained, but broken streaks can trigger guilt, shame, and all-or-nothing thinking.
- Memory (4/5): Daily reminders and visual history support habit memory, though the tool does not store contextual information or one-time tasks.
- Action (4/5): Promotes consistent daily action through clear prompts and routine reinforcement but lacks adaptability and nuance for changing circumstances.

2.2.9 Paper Planners

Overall, paper planners earned an average rating of 3.4 out of 5 (see Table C.9, Appendix C). They are highly effective for focus, mindfulness, and intentional planning, but lack smart features such as reminders, automation, and dynamic prioritization. They function best as an analog planning and reflection tool, rather than a comprehensive executive function support system for users who struggle with memory and task initiation.

- Activation (3/5): Physical presence and tactile interaction can prompt engagement, but planners can be forgotten, left behind, or feel intimidating when pages are blank.
- Focus (5/5): Completely free of digital distractions and encourages mindful, single-task planning, though it limits visibility across longer timeframes.
- Effort (3/5): Writing by hand supports memory and personalization, but updating plans is time-consuming and requires manual rewriting.
- Emotion (5/5): Personal, creative, and non-judgmental, offering tangible satisfaction without productivity pressure, though mistakes or messy pages can feel discouraging.
- Memory (2/5): Handwriting aids recall, but the lack of automatic reminders or alerts makes consistent checking necessary.

- Action (4/5): Provides clear task lists and satisfying completion but remains passive and cannot dynamically prioritize or prompt action.

2.3 What This Reveals

As stated before, many of these tools are marketed for students with executive function disorder, but none of them create a comprehensive system that addresses all six elements of executive function, or the full scope of their academic needs. Most tools prioritize memory and task tracking while neglecting **activation** and **emotional regulation**, offering little support for actually starting tasks or managing the shame, anxiety, or crisis states that often accompany academic work. These tools are optimized for feature richness, flexible customization, and disciplined, consistent use by power users, rather than for immediate usability, reduced activation energy, and forgiveness for inconsistency. As a result, critical gaps remain, including a lack of academic specificity (such as understanding semesters, syllabi, and assignment types), insufficient activation optimization, minimal emotional safety mechanisms, limited task breakdown support, and inadequate free access for students. Consequently, every existing tool contains at least one structural weakness that prevents it from offering truly comprehensive executive function support.

This comparative analysis reveals that current productivity tools consistently fail at activation and emotional regulation while optimizing for memory and tracking. Chapter 3 presents a design framework that inverts these priorities, specifically designed for executive dysfunction rather than adapting generic productivity systems.

Chapter 3: Website Development and Design Implementation

3.1 Design Philosophy and Foundational Principles

Before starting to design a website with the intention of helping students with executive dysfunction, one must have a deeper understanding of several psychological theories, models, and systems that would be used to create a website to better help them. This website uses Cognitive Load Theory, Universal Design for Learning (UDL), Self-Determination Theory, and the Fogg Model.

Cognitive Load Theory discusses how much information a person can take in at one time (Sweller, 1988; Medical College of Wisconsin, 2022). When designing a tool for students with executive dysfunction, steps need to be taken, so that information is presented at a rate best for each user to prevent cognitive overload which could lead to task abandonment. This can mean limiting the number of assignments one sees at a time or breaking down assignments into even more steps. Cognitive Load Theory also suggests that creating routine systems makes it easier for users to take in information.

Universal Design for Learning (UDL) is a set of guidelines based on the principle that different learners have different needs (Meyer, Rose, & Gordon, 2014; CAST, n.d.). Rather than designing standardized systems for an "average" user, UDL suggests that designing systems for someone who has the greatest needs creates a better product for all users. In fact, executive function is so important that one of the key updates from UDL 2.0 to UDL 3.0 was the addition of an entire category devoted to it. When designing an academic support tool, allowing users the choice to mark a task as "started," "paused," or "I need help", rather than only "complete", accommodates different planning styles and executive function strengths.

Self-Determination Theory explains motivation through three core psychological needs: autonomy, competence, and relatedness (Deci & Ryan, 2000). Students with executive dysfunction often experience repeated failure in traditional productivity systems, which can undermine their sense of competence and increase feelings of shame. When designing a tool for students with executive dysfunction, one must be careful of language choices, reduce factors that might trigger shame and instead promote a sense of power and agency (Deci & Ryan, 2000; Ryan & Deci, 2000).

The Fogg Model explains that, in order for a certain behavior to occur, a person must have sufficient motivation, ability, and a prompt or trigger (Fogg, 2009; Fogg Behavior Model, n.d.). When designing a tool for students with executive dysfunction, this perspective emphasizes that even small actions can help someone build momentum and support gradual progress.

The main design principle of this project was to design specifically for executive dysfunction rather than around it. Instead of designing solutions that address individual parts of executive dysfunction, the goal of this website is to create a well-rounded solution that addresses all parts of a student's academic life and reduce cognitive, emotional, and motivational barriers. Design decisions prioritize low activation effort to reduce barriers to starting tasks and treat emotional safety as a core requirement, minimizing shame through supportive language and non-punitive feedback. To reduce cognitive overload and context-switching, the system emphasizes integration over fragmentation and uses progressive disclosure to present only immediately relevant information while keeping additional complexity available when needed. Finally, the design follows a fail-safe approach, allowing users to disengage and return without data loss or penalty. These principles guide all technical and design decisions throughout implementation.

3.2 Core Features

3.2.1 Syllabus Processor

At the start of each semester, students face the overwhelming task of transferring all assignment due dates from multiple syllabi into their planning system. This high activation energy task often gets postponed, leaving students without a clear view of their semester workload until it's too late. The Syllabus Processor (see Figure D.1, Appendix D) allows students to upload their PDF syllabi, which is then automatically parsed to extract information like the professor's email, class meeting times, office hour times, grade breakdown by percentage, and all assignments, projects, and exams are extracted. Users can review the extracted information (see Figure D.2, Appendix D) and then send all the relevant dates and assignments to the other parts of the website.

This feature directly supports executive functioning by significantly reducing activation energy, strengthening memory, and enabling action. A process that typically takes several hours is reduced to a brief upload-and-review interaction, lowering the barrier to getting started and minimizing cognitive overload. This feature aligns with Cognitive Load Theory (Sweller, 1988) and the Fogg Model (Fogg, 2009) by taking a high-effort, high-friction task and turning it into a simple upload-and-review interaction, increasing user ability while minimizing activation barriers. In addition, automated extraction reduces the risk of missed or incorrectly recorded assignments, while immediate visibility into the full semester workload supports earlier planning and decision-making. The design choices in a feature like this are aimed to prioritize usability and reduce anxiety during interaction.

3.2.2 Integrated Calendar

Traditional digital calendars often present all commitments at once, resulting in visual overload that can leave students feeling paralyzed rather than informed. While tools like Google Calendar effectively show *when* something is due, they provide little support for understanding *what* needs to be done or *how* to approach tasks. The integrated calendar (see Figure D.4, Appendix D) in this system is designed to reduce that visual overload by offering multiple filtered views tailored to different planning needs, including a Today view, adjustable Week and Month views, a Deadline-only view, and a Focus Mode that hides everything except the current task. By automatically syncing with the Syllabus Processor and user-entered tasks, the calendar creates a single, cohesive timeline without requiring duplicate data entry.

This calendar supports executive function by strengthening memory through clear visual representations of time commitments, improving focus through selective information filtering, and supporting emotional regulation through customizable visibility settings. Features such as time blocking, course-based color coding, collision detection, buffer-time suggestions, and a "Crisis Mode" calendar help students plan realistically while maintaining flexibility. Design and language choices, prioritize emotional safety and adaptability, ensuring the calendar supports students during both routine planning and periods of overwhelm. The calendar's adaptive structure reflects Cognitive Load Theory (Medical College of Wisconsin, 2022) and Universal Design for Learning (Meyer, Rose, & Gordon, 2014), ensuring students only see information that is immediately relevant while retaining access to broader context when needed.

3.2.3 Priority Task List

Students with executive dysfunction often struggle most not with knowing *what* needs to be done, but with deciding *where to start*. Traditional to-do lists present all tasks with equal visual weight, placing the burden of prioritization entirely on the user and often triggering decision paralysis. The Priority Task List (see Figure D.3, Appendix D) addresses this gap by using an intelligent prioritization system that surfaces the most relevant and actionable tasks based on urgency, estimated completion time, energy level, course workload balance, task dependencies, and past user behavior. Instead of confronting users with a long list, the system presents a clear "Next 3 Tasks" view that directly answers the question, *"What should I do right now?"*

This feature strongly supports executive function by reducing activation barriers, clarifying immediate actions, and lowering anxiety caused by overwhelming choice. One-click task start, energy and time-based filtering, and a quick-capture inbox minimize friction and allow students to engage at their current capacity. Completed tasks automatically hide to reduce visual clutter and shame, while non-judgmental language and optional celebration animations support emotional safety. Design choices such as how users would like to see their progress, and how they would like to have things prioritized ensure the system remains flexible, forgiving, and responsive, which guides students into action without requiring constant self-regulation or complex decision-making.

3.2.4 Study Buddy Feature: Gamification Without Guilt

Traditional gamification systems often rely on streaks, points, and visible performance metrics that can create pressure, guilt, and fear of failure for anyone, let alone students with executive

dysfunction. When progress is framed around consistency or loss, a missed day can feel like personal failure rather than a small setback. The Study Buddy feature (see Figure D.6, Appendix D) is intentionally designed as a *minor, optional* component that offers encouragement without obligation or punishment. It exists in a separate tab and never intrudes into core task management workflows, allowing users the ability to fully ignore it without losing any functionality or support elsewhere in the system.

The Study Buddy provides gentle, positive reinforcement by tracking only completed actions such as finishing tasks, starting focus sessions, or breaking down assignments. While points are awarded, they are used solely for cosmetic or personalization purposes. Users can unlock new colors and patterns for the website, font styles, or customizations for their Study Buddy. There are no streaks, leaderboards, penalties, or productivity comparisons. Points are still awarded in crisis mode or if users ask for help. The goal of the Study Buddy is to add another form of accountability and fun, not to add negative pressure to users. This feature is grounded in the concepts of Self-Determination Theory (Deci & Ryan, 2000) and the Fogg Behavior Model (Fogg, 2009), since it creates additional user-driven motivation and lowers barriers to activation through encouragement. Supportive messages reinforce effort rather than output, emphasizing emotional safety and self-compassion. By keeping gamification quiet, optional, and customizable, the Study Buddy supports emotion, effort, and focus without becoming a source of pressure, distraction, or shame.

3.2.5 Assignment Breakdown

The AI Assignment Breakdown feature (see Figure D.5, Appendix D) addresses one of the most common barriers for students with executive dysfunction: not knowing how to approach large assignments. Instead of staring at a vague task like "Write a paper on a policy" and feeling stuck, students can click the "Break This Down" button to receive detailed, step-by-step instruction generated by AI directly applying principles from Cognitive Load Theory (Sweller, 1988). Each step includes realistic time estimates, suggested order, and actionable guidelines, with options to edit, add, or remove steps as needed. The system also adds these new sub-tasks to the user's calendar and to-do list, keeping each step smaller and separate but also still contributing to the finished assignment. By offloading the cognitive effort of this would require to AI, the system directly supports activation, effort and emotion, providing a clear set of instructions from start to finish while preserving the student's mental energy to complete the actual work.

This feature strongly supports executive function by eliminating activation barriers, clarifying action, and preserving mental energy for actual work rather than planning. Activation improves because the question of "how do I start" is answered. Action is supported since the user is given clear ways to do the assignment. Effort is supported since the extra motivation needed for organizational thinking is offloaded. This feature is like the handrails on a staircase for students with executive dysfunction, providing an extra level of support and emotional safety while maintaining flexibility for the individual.

3.2.6 Crisis Mode

When a student falls behind, they can experience shame or embarrassment. This is only worsened for students with executive dysfunction who have difficulty with emotional regulation. These feelings can prevent them from opening any productivity tools at all, creating a cycle of avoidance that only gets worse. Crisis Mode addresses this by providing a psychologically safe reset that aligns with Cognitive Load Theory (Medical College of Wisconsin, 2022). Temporarily, extraneous emotional and informational load is reduced so students can re-engage. A prominent "Feeling Overwhelmed?" button activates Crisis Mode, which hides overdue tasks, filters the system to only what must be completed in the next 48 hours, and prompts the student with a single, manageable question about immediate priorities. This temporary solution helps students focus on not falling further behind by working on already late assignments or future assignments. Depending on the severity of the crisis, it gets them back on track and then works with the student to address whatever went wrong or helps them get the external help they need.

Crisis Mode strongly supports executive function by prioritizing emotion regulation and activation before productivity. It reduces shame by removing visual reminders of missed work, supports activation by narrowing focus to one small, achievable action, and supports action by creating a clear, step-by-step recovery path. Built-in communication templates help students reach out to professors, advisors, disability services, or peers without the added cognitive burden of drafting difficult messages. Design choices ensure that Crisis Mode functions as a fail-safe recovery mechanism rather than a punitive feature, but still is a welcoming tool. Crisis Mode prioritizes getting students back on track over teaching long-term skills, based on the principle that someone in crisis cannot simultaneously manage burnout and build new coping strategies.

Like emergency interventions in psychology, the goal is to stabilize first, then address underlying patterns.

3.3 Design Decisions

The design decisions in this project were intentionally driven by known barriers to executive function in students. Each feature was evaluated through the lens of activation, focus, effort, memory, action, and emotion, ensuring that no part of the system required students to "push through" cognitive or emotional difficulties that could lead to tool abandonment. Low activation effort was prioritized across the platform through limited clicks, automated setup, smart defaults, and AI-assisted planning. These choices reduce the cognitive burden required to begin tasks, directly addressing initiation paralysis that is common among students with executive dysfunction. Rather than expecting users to organize, prioritize, and plan independently, the system provides a structure that adapts to a potential user's varying levels of capacity and energy.

A central design goal of this system is to replace the need for multiple, disconnected productivity tools. Students with executive dysfunction are often required to juggle calendars, to-do lists, note apps, learning management systems, and communication platforms, creating constant context-switching and cognitive overload (Ali & Thu, 2025). This website consolidates those functions into a single, integrated environment that supports the full academic workflow, from syllabus intake and semester planning to task execution, crisis recovery, and communication. By eliminating fragmentation, the system reduces app fatigue, prevents information loss between platforms, and allows students to maintain a coherent mental model of their academic responsibilities without repeatedly re-orienting themselves.

Many of these decisions directly respond to gaps identified in the comparative analysis of existing productivity tools. High setup barriers found in platforms like Notion are addressed through the Syllabus Processor's easy semester setup. Shame-inducing features such as prominently displayed overdue tasks or streak-based punishments are replaced with Crisis Mode and an optional Study Buddy. Similarly, the absence of task breakdown guidance in traditional calendars is addressed through AI-generated assignment plans that translate vague expectations into concrete steps. Rather than forcing students to adapt to rigid systems, the design for this tool prioritizes flexibility and emotional safety for its users. This ensures the tool supports students consistently across varying levels of motivation, focus, and stress without requiring multiple external apps.

Chapter 4: Conclusions, Implications, and Future Directions

4.1 Summary of Key Findings

This capstone project examined **two interconnected research questions** focused on the effectiveness of existing productivity tools for students with executive dysfunction. Specifically, the study explored which features of current tools help or hinder students with executive dysfunction and which executive function components are most consistently neglected in existing designs. The findings demonstrate that shortcomings in these tools are not isolated or accidental, but rather reflect systematic design patterns that fail to account for the cognitive, emotional, and activation-related challenges faced by neurodivergent students. Together, these results highlight both the limitations of current productivity technologies and the clear opportunities for more inclusive, executive-function-informed design.

First, comparative analysis of nine widely used productivity tools revealed an imbalance in executive function support. Memory-related features such as reminders and notifications were the strongest supported component (average 3.9/5), while Activation and Emotion were consistently the weakest (both averaging 2.9/5). This disparity reflects a design bias toward technically straightforward solutions rather than psychologically informed ones. While reminders are easy to implement, supporting task initiation and emotional regulation requires deliberate design choices that prioritize user safety, simplicity, and encouragement.

Second, no single tool effectively supports all six executive function components. Currently users must pick and choose what features they find important. They could choose a simple tool with low activation barriers but lack planning depth, or a feature-rich platform that has more features and flexibility but has an overwhelming setup and a higher potential for cognitive

overload. They could also choose a focus tool during work sessions, but it does not have the ability to help with planning. AI tools reduce cognitive effort without supporting execution or follow-through. Even tools advertised as "all-in-one" fail to integrate academic context or address emotional barriers.

Third, there are specific design features that helped or hindered students with executive dysfunction. Features such as voice input, automatic saving, one-click task capture, clear next-action visibility, and non-judgmental language consistently reduce activation effort and anxiety. On the other hand, complex setups, rigid organizational structures, productivity metrics, overdue task displays, and feature overload increase user avoidance and shame. These findings show that executive dysfunction is not incompatible with productivity tools, but rather with common design practices.

Finally, academic context is largely absent from generic productivity tools. College work operates within predictable structures. Each semesters students get their syllabi, they can see their assignments, exam schedule, and other deadlines. Most tools still treat academic tasks as interchangeable with general to-do items. The lack of syllabus integration, assignment breakdown guidance, and course-specific organization represents a significant and addressable design gap for students who rely on additional structure to function effectively.

Collectively, these findings support the conclusion that effective tools for executive dysfunction require **design-first rather than feature-first thinking**. Students do not need their productivity tools to have more features. They need systems that reduce activation-energy, minimize cognitive load, ensure emotional safety, and meet them where they are.

4.2 Future Directions

While design decisions were grounded in existing research and comparative analysis, the tool's effectiveness remains theoretical rather than empirically validated. Future testing is essential to confirm the accuracy of assumptions, identify unanticipated barriers, and evaluate real-world impact. If this project does in fact continue, it should begin with structured user testing across college students with executive dysfunction to validate usability, emotional impact, and activation speed.

Medium-term development would focus on strengthening the tool's technical infrastructure and expanding core functionality. Priorities would include optional cloud syncing to support cross-device use more sophisticated and customizable reminder systems, assignment templates adaptable to different academic tasks, course-specific workspaces, integrated audio-based focus supports, and optional body-doubling features. These enhancements would deepen executive function support while maintaining low activation barriers.

In the long term, this tool could move beyond functioning solely as a "helping" system and instead serve as a learning platform where students can develop transferable skills that can help them understand and eventually learn systems to minimize their executive function challenges. Rather than simply being a platform to help students with their difficulties the system could gradually help students to building awareness of their difficulties with these processes and help them practice strategies that are necessary beyond academic tasks. This skill-building approach would remain optional and supportive, avoiding pressure toward premature independence while allowing users to engage with reflective prompts, strategy explanations, and adaptive support as

needed. This vision would be reinforced through institutional partnerships, including integration with learning management systems and collaboration with campus disability services, reducing the need for manual data entry and positioning the tool as a legitimate accommodation. Additional partnerships with complementary executive function support, such as academic coaching programs and evidence-based focus or audio services, would further strengthen the ecosystem of support while preserving a unified, low-friction user experience.

4.3 Broader Implications

This research challenges the prevailing feature-first approach that is used in educational technology. Designing for executive dysfunction reveals that many common productivity features actively harm usability for a significant population of students. Importantly, features that support executive dysfunction like non-judgmental interfaces, clear next-action guidance, and reduced setup burden actually benefit all users.

At an institutional level, these findings support treating productivity tools as legitimate academic accommodations and integrating executive function support directly into educational infrastructure. By reframing executive dysfunction as a design issue rather than a personal failure, this would help reduce stigmas and create more inclusive technology practices.

4.4 Final Reflections

This capstone began with personal frustration: Why don't productivity tools work for me? The journey from that question to this comprehensive analysis has revealed that the answer is both simple and complex.

Simply, current tools aren't designed with executive dysfunction in mind. They're designed for people who can easily start tasks, maintain focus, and handle burn-out, the very things executive dysfunction makes difficult.

Designing tools that actually work requires deeper understanding of executive function theory (Barkley, 2012; Brown, 2013; CHADD, n.d.), careful attention to emotional impact, willingness to prioritize simplicity over features, and recognition that neurodivergent users have different needs that can't be bolted on as accommodations.

The website prototype (https://lhorowi.github.io/capstone/) demonstrates that comprehensive executive function support is achievable. Whether through commercial development, open-source collaboration, or inspiring others to build better tools, this work provides a foundation for change.

The ultimate goal isn't just to create one good tool. It's to change how we think about designing technology for neurodivergent users. Instead of asking "How can we accommodate executive dysfunction?" we should ask "How can we design for executive function differences from the start?"

Having executive dysfunction does not mean that you are incompetent. It means you need different structures and supports. Technology should remove barriers, not create them. This capstone is one step toward that future.

Appendices

Appendix A: Literature Review Source Summary

Foundational Understanding of Executive Function

1. Cleveland Clinic. (n.d.). Executive function. https://my.clevelandclinic.org/health/articles/executive-function

- Three core areas of executive function
- Medical/clinical definition and explanation
- Practical symptoms and manifestations

2. Barkley, R. A. (2012). *Executive functions: What they are, how they work, and why they evolved.* Guilford Press.

- Four-component model of executive function
- Self-regulation as core deficit in ADHD
- Neurological and evolutionary basis for executive dysfunction

3. Brown, T. E. (2013). *A new understanding of ADHD in children and adults: Executive function impairments.* Routledge.

- Six-cluster model of executive function
- Integration of clusters in ADHD
- Clinical implications for intervention

4. CHADD. (n.d.). Executive function skills. https://chadd.org/about-adhd/executive-function-skills/

- Comparison of Barkley and Brown models
- Practical applications for students
- Connection between EF and ADHD symptoms

5. Diamond, A. (2013). Executive functions. *Annual Review of Psychology*, *64*, 135-168. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4084861/

- Comprehensive review of executive function research
- Developmental trajectory of EF skills
- Neural mechanisms and interventions

6. How to Teach Executive Functioning Skills. (2020, July 29). [Video]. YouTube. https://www.youtube.com/watch?v=H4oIQrBe8aI

- Practical executive function skills for students
- Three self-coaching prompts
- Application strategies for daily life

7. Branching Minds. (n.d.). Executive function toolkit. https://www.branchingminds.com/blog/executive-function-toolkit

- Teaching strategies for students with executive function challenges
- Practical interventions for educators
- Classroom implementation guidance

8. Understood.org. (n.d.). Classroom accommodations for executive function challenges. https://www.understood.org/en/articles/classroom-accommodations-executive-function-challenges

- Specific accommodations for EF challenges
- Evidence-based support strategies
- Implementation in educational settings

9. Rapport, M. D., Orban, S. A., Kofler, M. J., & Friedman, L. M. (2013). Do programs designed to train working memory, other executive functions, and attention benefit children with ADHD? A meta-analytic review of cognitive, academic, and behavioral outcomes. *Clinical Psychology Review*, *33*(8), 1237-1252. https://pmc.ncbi.nlm.nih.gov/articles/PMC3758442/

- Meta-analysis of executive function training programs
- Evidence for intervention effectiveness

- Cognitive training outcomes for ADHD

ADHD and College Students

10. DuPaul, G. J., Weyandt, L. L., O'Dell, S. M., & Varejao, M. (2009). College students with ADHD: Current status and future directions. *Journal of Attention Disorders*, *13*(3), 234-250. https://doi.org/10.1177/1087054709340650

- Prevalence and impact of ADHD in higher education
- Academic performance patterns
- Recommendations for support services

11. Haro, J. M., et al. (2022). ADHD and academic performance in college students: A systematic review. *Journal of Attention Disorders*, *26*(1). https://journals.sagepub.com/doi/10.1177/10870547241306554

- Systematic review of ADHD academic outcomes
- GPA and graduation rate data
- Factors affecting academic success

12. Sedgwick, J. A., Merwood, A., & Asherson, P. (2022). The relationship between executive function, neurodevelopmental disorder traits, and academic achievement in university students. *Frontiers in Psychology*, *13*. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.958013/full

- EF as predictor of academic achievement
- Working memory and university performance
- Neurodevelopmental traits in college populations

13. Prevatt, F., & Young, J. L. (2014). Recognizing and treating attention-deficit/hyperactivity disorder in college students. *Journal of College Student Psychotherapy*, *28*(3), 182-200.

- Recognition and diagnosis in college settings
- Treatment approaches for college students
- Psychotherapy and coaching strategies

14. Hinshaw, S. P., & Scheffler, R. M. (2014). *The ADHD explosion: Myths, medication, money, and today's push for performance.* Oxford University Press.

- Social context of ADHD diagnosis
- Medication and treatment landscape
- Cultural factors in ADHD prevalence
- Academic Coaching Research

15. Campbell, A. L., & Mogashana, D. (2024). Assessing the effectiveness of academic coaching interventions for student success in higher education: A systematic review. *Innovations in Education and Teaching International*. https://doi.org/10.1080/14703297.2024.2417173

- Systematic review of coaching effectiveness
- Evidence for collaborative coaching models
- Cost and scalability challenges in higher education

16. Syracuse University Center for Disability Resources. (n.d.). OnTrack at SU: Program overview. https://disabilityresources.syr.edu/ontrackatsu/program-overview/

Example institutional coaching program

- Cost structure and program details
- Scalability and access limitations
- App Fatigue and Digital Tools

17. Ali, S. H., & Thu, H. (2025). App fatigue in mHealth: Beyond improving apps, advance equity by meeting people where they are. *PLOS Digital Health*, *4*(11), e0001107. https://doi.org/10.1371/journal.pdig.0001107

- Definition of app fatigue and app overload
- Cognitive burden of managing multiple systems
- Structural problems beyond app quality

Need for integrated, embedded solutions

18. Author information unavailable. (n.d.). Student technology use and preferences [PDF]. https://drive.google.com/file/d/1JZFd3yTXXUbNZQpsjlG_UrXsTUZUM9Yl/view

- Student technology adoption patterns
- Tool preferences and usage data
- Context for productivity tool selection

19. Author information unavailable. (n.d.). Technology tools and student productivity [Research article]. https://pmc.ncbi.nlm.nih.gov/articles/PMC6406620/

- Digital tool usage in educational contexts
- Impact on student outcomes
- Technology integration challenges

20. Author information unavailable. (n.d.). Which productivity tools are popular: A study. *ERIC*, EJ1353516. https://files.eric.ed.gov/fulltext/EJ1353516.pdf

- Survey of productivity tool popularity among students
- Usage patterns and preferences
- Comparative analysis data
- Design Theory and Frameworks

21. Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, *12*(2), 257-285.

- Original cognitive load theory
- Three types of cognitive load (intrinsic, extraneous, germane)
- Implications for instructional design
- Working memory limitations

22. Sweller, J., Ayres, P., & Kalyuga, S. (2011). *Cognitive load theory.* Springer.

- Comprehensive treatment of CLT
- Instructional design principles
- Application to educational technology

23. Medical College of Wisconsin. (2022). Cognitive load theory: A guide to applying cognitive load theory to your teaching [PDF]. https://www.mcw.edu/-/media/MCW/Education/Academic-Affairs/OEI/Faculty-Quick-Guides/Cognitive-Load-Theory.pdf

- Practical application guide
- Teaching strategies based on CLT
- Managing cognitive load in instruction

24. Deci, E. L., & Ryan, R. M. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, *55*(1), 68-78.

- Three psychological needs: autonomy, competence, relatedness
- Intrinsic vs extrinsic motivation
- Applications to educational contexts
- Implications for tool design

25. Ryan, R. M., & Deci, E. L. (2000). The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry*, *11*(4), 227-268.

- Goal pursuit and need satisfaction
- Behavioral self-determination
- Motivational processes

26. Self-Determination Theory. (n.d.). Theory. https://selfdeterminationtheory.org/theory/

- Official SDT framework documentation
- Research applications across domains
- Mini-theories and components

27. Fogg, B. J. (2009). A behavior model for persuasive design. *Proceedings of the 4th International Conference on Persuasive Technology*, 1-7.
- B=MAP formula (Behavior = Motivation × Ability × Prompt)
- Design implications for behavior change
- Application to product design

28. Fogg, B. J. (2020). *Tiny habits: The small changes that change everything.* Houghton Mifflin Harcourt.
- Practical application of behavior model
- Habit formation strategies
- Behavioral design methodology

29. Fogg Behavior Model. (n.d.). Behavior model. https://www.behaviormodel.org/
- Official Fogg Behavior Model documentation
- Interactive resources and tools
- Design applications

30. Meyer, A., Rose, D. H., & Gordon, D. (2014). *Universal Design for Learning: Theory and practice.* CAST Professional Publishing.
- Three UDL principles
- Design for variability
- Multiple means of engagement, representation, and action
- Practical implementation strategies

31. CAST. (n.d.). UDL guidelines. https://udlguidelines.cast.org/
- Complete UDL Guidelines framework
- Interactive guideline explorer
- Checkpoints and implementation examples
- Updates and version history

32. Goodwin University. (n.d.). Universal Design for Learning (UDL): 9 in 9 webinar series [Video playlist]. YouTube. https://youtube.com/playlist?list=PL4adNvqC2hbwu19Q-dIb8FZB_Ic5ZgX_S
- Video instruction on UDL implementation
- Practical examples and case studies
- Nine-part instructional series
- Productivity Tools Analyzed

33. Google LLC. (n.d.). Google Calendar: Online calendar and scheduling tool. https://calendar.google.com

34. Notion Labs, Inc. (n.d.). Notion: All-in-one workspace for notes, tasks, and projects. https://www.notion.so

35. Doist Inc. (n.d.). Todoist. https://todoist.com

36. Atlassian Pty Ltd. (n.d.). Trello. https://trello.com

37. Apple Inc. (n.d.). Reminders. https://www.apple.com/ios/reminders/

38. Focus To-Do Team. (n.d.). Focus To-Do: Pomodoro timer & tasks. https://www.focustodo.cn/

39. OpenAI. (n.d.). ChatGPT. https://chat.openai.com

40. HabitRPG, Inc. (n.d.). Habitica. https://habitica.com

Appendix B: Executive Function Evaluation Framework

Table B.1: Evaluation Criteria for Productivity Tools
This framework was used to systematically assess each of the nine productivity tools across executive function components.

Component	What to Look For	Rating Scale
Activation	Does the tool help users *get started* on tasks? Measures: ease of initial setup, time to first task entry, required fields, decision-making burden	1 (Poor) to 5 (Excellent)
Focus	Does it help minimize distractions or sustain attention? Measures: visual clarity, filtering options, notification management	1 (Poor) to 5 (Excellent)
Effort	Does it support maintaining effort and motivation over time? Measures: maintenance burden, automation features, ongoing cognitive load	1 (Poor) to 5 (Excellent)
Emotion	Does it reduce stress, anxiety, or frustration related to tasks? Measures: language tone, handling of overdue items, shame triggers	1 (Poor) to 5 (Excellent)
Memory	Does it help capture and organize information for later use? Measures: reminder systems, notification options, information persistence	1 (Poor) to 5 (Excellent)
Action	Does it support task completion and self-monitoring? Measures: task breakdown support, next-action clarity, progress tracking	1 (Poor) to 5 (Excellent)
Usability	Is it practical for college students' academic needs? Overall ease of use and learning curve	1 (Poor) to 5 (Excellent)
Effectiveness	General impression of how well it supports executive function challenges in academic contexts	1 (Poor) to 5 (Excellent)
Accessibility	Is the tool truly free, easy to use, and available across devices?	1 (Poor) to 5 (Excellent)

Rating Definitions:
1: Actively hinders this function
2: Minimal or ineffective support
3: Basic support with notable limitations
4: Strong support with minor gaps
5: Exceptional support for this component

Appendix C: Detailed Tool Evaluation Tables

Table C.1: Google Calendar

Component	Rating	Examples	Pros	Cons
Activation	4	Quick event creation, voice input, automatic time suggestions	Easy to get started, low barrier to entry, familiar interface	Can feel overwhelming with multiple calendars, initial setup requires decisions
Focus	**3**	Color-coding, day/week/month views, "Focus time" blocks	Visual separation of tasks, can block distracting time slots	Too many events create visual clutter, notifications can be distracting
Effort	**4**	Drag-and-drop rescheduling, repeat	Minimal effort to maintain once set	Requires consistent input, easy to ignore or dismiss

Component	Rating	Examples	Pros	Cons
		events, time estimates built-in	up, automatic syncing	
Emotion	**3**	Neutral interface, customizable colors	Non-judgmental, doesn't create guilt	Can feel overwhelming when overscheduled, no encouragement/motivation
Memory	**5**	Multiple reminder options, email/push notifications, location-based reminders	Excellent external memory support, persistent reminders	Over-reliance can lead to notification fatigue
Action	**4**	Clear time-blocking, integration with other apps, one-click join for meetings	Shows exactly when to do things, reduces decision-making	Doesn't break tasks into steps, passive (doesn't prompt action beyond reminders)
Usability	**4**	Intuitive for most users, cross-platform, widely taught in schools		
Effectiveness	**4**	Strong for time-based commitments, excellent for appointments and deadlines		
Accessibility	**5**	Free, screen reader compatible, works on all devices, voice control available		

Table C.2: Notion

Component	Rating	Examples	Pros	Cons
Activation	2	Blank page syndrome, requires setup/templates, complex database creation	Powerful once configured, templates available	Extremely high activation energy, overwhelming for beginners, too many options
Focus	2	Can create focused pages, toggle lists, minimal mode available	Customizable to reduce distractions, can hide completed items	Endless scrolling, easy to get lost in pages, sidebar clutter, too much flexibility
Effort	2	Requires manual organization, database maintenance, linking pages	Once set up can be low-effort, everything in one place	High ongoing maintenance, easy to create messy systems, time-consuming to organize
Emotion	2	Highly customizable aesthetics, can be beautiful	Can create motivating workspace, personal expression	Perfectionism trap, comparison with others' setups, guilt over unused complexity
Memory	3	Database views, linked pages, can embed reminders	Everything stored in one place, searchable	No native robust reminder system, easy to forget to check, buried information

Component	Rating	Examples	Pros	Cons
Action	2	Checkboxes, templates, linked databases	Can create action-oriented workflows	Doesn't prompt action, requires going to Notion, too many steps to complete tasks
Usability	2	Steep learning curve, complex interface, frequent updates change features		
Effectiveness	3	Can be highly effective IF properly set up, but setup is the challenge		
Accessibility	3	Free tier available, improving accessibility features, but still limited screen reader support		

Table C.3: Todoist (“Do Everything Tools”)

Component	Rating	Examples	Pros	Cons
Activation	4	Quick add, natural language input, voice entry	Fast task capture, low friction to start	Can become dumping ground, initial project setup required
Focus	4	Today view, filters, priority levels, "focus" modes	Clear "what's next" view, hides future tasks	Can still feel overwhelming with large lists, priority paralysis
Effort	4	Recurring tasks, templates, quick rescheduling, one-click complete	Low ongoing effort, fast interactions	Requires regular review/cleanup, easy to accumulate overdue tasks
Emotion	3	Karma/productivity scores (Todoist), streaks, neutral design	Gamification can motivate, satisfaction from checking off	Guilt from overdue tasks, stress from red notifications, productivity shame
Memory	5	Persistent reminders, location-based reminders, smart date parsing	Excellent external memory, reduces mental load	Notification fatigue possible, can over-rely
Action	4	Clear next actions, priority sorting, time blocking (TickTick)	Action-oriented design, reduces decision fatigue	Doesn't break down tasks automatically, requires self-discipline
Usability	5	Highly intuitive, clean interface, cross-platform consistency		
Effectiveness	4	Very effective for task management, strong for students with EF challenges		
Accessibility	4	Affordable/free tiers, keyboard shortcuts, voice input, decent screen reader support		

Table C.4: Trello

Component	Rating	Examples	Pros	Cons
Activation	3	Quick card creation, templates available, drag-and-drop	Visual and interactive, easy to add items	Requires board setup, can feel overwhelming to start, decision about structure
Focus	3	Board views, can focus on single list, filters	Visual organization, can see workflow stages	Too many cards create clutter, endless scrolling, multiple boards to check
Effort	3	Drag-and-drop simplicity, automation (Butler in Trello), templates	Visual progress feels rewarding, easy to reorganize	Requires maintaining board organization, cards accumulate, manual updates needed
Emotion	4	Visual progress (moving cards), satisfying interactions, customizable	Sense of accomplishment, less judgmental than lists	Can feel overwhelming with large projects, abandoned boards create guilt
Memory	3	Due date reminders, card comments, checklists	Good for project details, persistent visual cues	Not optimized for daily reminders, easy to miss due dates, requires checking
Action	3	Checklists within cards, clear workflow stages	Good for breaking down projects, shows next stage	Doesn't prompt specific actions, requires self-initiation, multi-step process to complete
Usability	4	Intuitive visual interface, but can become complex with many features		
Effectiveness	3	Better for project management than daily task execution, good for visual thinkers		
Accessibility	4	Free tiers available, improving keyboard navigation, screen reader support varies		

[Tables C.5 through C.9 would continue with Phone Reminders, Pomodoro Apps, ChatGPT, Habit Trackers, and Paper Planners following the same format]

Table C.5: Phone Reminders (iOS)

Component	Rating	Examples	Pros	Cons
Activation	5	Voice commands ("Hey Siri, remind me..."), always accessible, one-tap creation	Lowest possible friction, always available, instant capture	None significant for activation
Focus	4	Simple lists, today view, minimal interface	No feature bloat, distraction-free	Limited organization options, can't see the big picture
Effort	5	Minimal features means minimal effort, automatic syncing, swipe to complete	Extremely low effort, no setup needed	Limited functionality may require other tools

Component	Rating	Examples	Pros	Cons
Emotion	4	Non-judgmental, simple completion, no tracking/guilt	Stress-free, no performance pressure	No motivation/encouragement, boring interface
Memory	5	Persistent notifications, location-based reminders, time-based alerts	Excellent for external memory, hard to ignore	Can create notification overload
Action	4	Clear single actions, immediate notifications, location triggers	Prompts action effectively, simple one-step completion	Can't break down complex tasks, no subtasks (iOS basic version)
Usability	5	Extremely simple, pre-installed, no learning curve		
Effectiveness	4	Highly effective for simple reminders and immediate tasks, less so for complex planning		
Accessibility	5	Free, built-in, voice control, excellent screen reader support		

Table C.6: Pomodoro Apps (Focus To-Do)

Component	Rating	Examples	Pros	Cons
Activation	4	One-tap to start, no setup required, visual cues (growing tree)	Low barrier to begin work, gamified start	Requires initiating the session, timing decisions can delay start
Focus	5	Forced focus periods, blocks distracting apps, visual timer	Excellent for sustained attention, external accountability	Can feel restrictive, breaking timer feels like failure
Effort	4	Simple start/stop, automatic breaks, minimal configuration	Low cognitive load during use	Requires discipline to start, manual timer restarts
Emotion	4	Rewards (growing trees), break reminders, sense of accomplishment	Positive reinforcement, breaks reduce burnout	Guilt if timer interrupted, pressure to complete full sessions
Memory	3	Session history, statistics, streak tracking	Shows work patterns, motivating history	Doesn't remind you to start sessions, no task reminders
Action	5	Forces immediate action, structures work time, mandatory breaks	Excellent for initiating and sustaining work, reduces burnout	Only helps during session, doesn't organize what to do
Usability	5	Very simple interface, clear purpose, intuitive		

Component	Rating	Examples	Pros	Cons
Effectiveness	5	Highly effective for focus and work initiation, excellent for EF challenges		
Accessibility	4	Many free options, visual and audio cues, simple interface (some apps lack screen reader support)		

Table C.7: AI Tools (ChatGPT)

Component	Rating	Examples	Pros	Cons
Activation	5	Conversational interface, no "right way" to start, forgiving	Zero intimidation, can ask anything, immediate response	Requires internet, can get distracted in conversation
Focus	3	One-on-one conversation, focused responses	Eliminates choice paralysis, guided interaction	Can go down rabbit holes, tangential conversations
Effort	5	AI does the heavy lifting, breaks down tasks, generates plans	Massively reduces cognitive load, creates structure	Requires asking, doesn't track long-term, need to implement suggestions
Emotion	5	Non-judgmental, encouraging, patient, reframes challenges	Reduces shame and anxiety, motivating, supportive	Not human connection, can feel impersonal
Memory	2	Conversation history, can create reminders in other apps	Remembers context in conversation	Doesn't actively remind you, need to return to check, no notifications
Action	4	Creates actionable steps, templates, guides through process	Excellent for planning and breaking down tasks	Doesn't track completion, requires external tools for execution
Usability	5	Natural conversation, no learning curve, accessible		
Effectiveness	4	Excellent for planning and problem-solving, less effective for ongoing tracking		
Accessibility	4	Free tier available (ChatGPT), text-based accessibility, works with screen readers		

Table C.8: Habit Trackers (Habitica)

Component	Rating	Examples	Pros	Cons
Activation	3	Daily prompts, notifications, visual cues	Reminders to engage, gamification motivates start	Requires daily check-in, easy to forget, initial setup decisions
Focus	4	One habit at a time, clear daily view, limited choices	Reduces overwhelm, clear daily goals	Can feel restrictive with multiple habits

Component	Rating	Examples	Pros	Cons
Effort	4	One-tap check-off, simple interface, automatic tracking	Very low effort to maintain, satisfying interaction	Requires consistent logging, manual input
Emotion	*	Visual progress, streaks, rewards (Habitica), celebrations	Highly motivating, dopamine hits, sense of achievement	Guilt from broken streaks, anxiety about maintaining, can feel like failure
Memory	4	Daily reminders, streak counters, visual history	Good external memory for habits, persistent nudging	Only tracks habits, not one-time tasks
Action	4	Clear daily checklist, specific habit prompts	Promotes consistent action, reduces decision-making	Passive tracking, doesn't prevent skipping, binary (done/not done)
Usability	4	Simple interfaces, gamification makes it engaging		
Effectiveness	4	Very effective for building routines, excellent for consistency		
Accessibility	4	Affordable, mobile-optimized, simple interactions (screen reader support varies)		

***** Habit trackers receive a paradoxical rating (5/1) because they are extremely motivating when streaks are maintained but create intense guilt and shame when streaks are broken, often leading to permanent tool abandonment.

Table C.9: Paper Planners

Component	Rating	Examples	Pros	Cons
Activation	3	Physical presence as reminder, no tech barriers, tactile	No login/loading, always accessible, visible on desk	Can be forgotten/left at home, requires finding/opening
Focus	5	No digital distractions, single-purpose, limited space forces prioritization	Zero distractions, mindful planning, can't multitask	Can be limiting, can't easily see long-term view
Effort	3	Handwriting is slow, manual transfer, no automation	Mindful and memorable, no technical issues	Time-consuming, no reminders, manual rewriting, mistakes permanent
Emotion	5	Satisfying to cross off, personal expression, no metrics/judgment	Creative outlet, no productivity shame, tangible progress	Messy pages can be discouraging, mistakes visible, can feel wasteful
Memory	2	No automatic reminders, visual presence only	Handwriting aids memory, visible reference	Easy to forget to check, no alerts, can be lost/damaged
Action	4	Physical act of crossing off satisfying, clear list format	Tangible completion, structured approach	Doesn't prompt action, passive, requires checking

Component	Rating	Examples	Pros	Cons
Usability	4	No learning curve, intuitive, portable		
Effectiveness	3	Can be very effective for those who prefer analog, varies by individual		
Accessibility	3	One-time purchase, no tech required, but requires physical dexterity and vision		

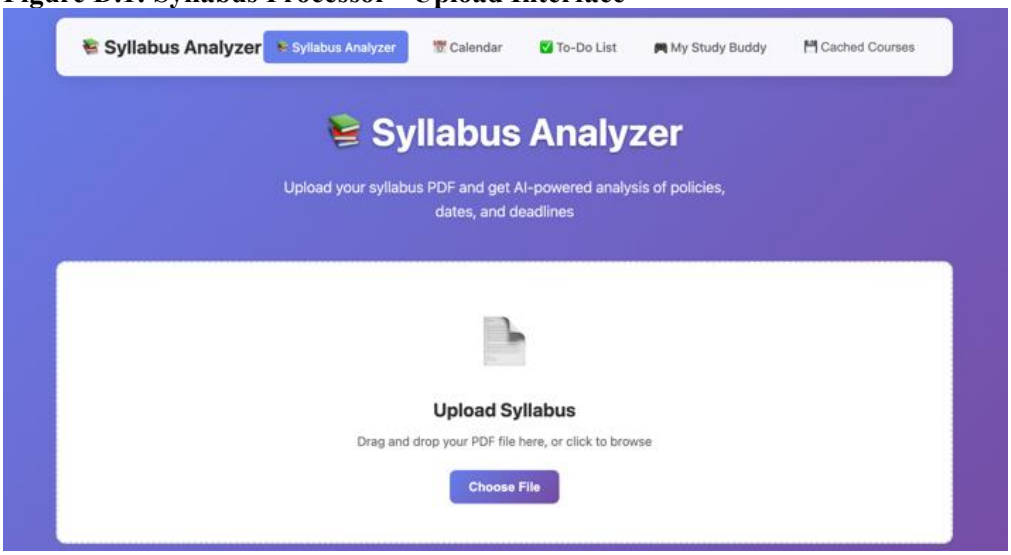
Table C.10: Aggregate Findings Across All Tools

Component	Average Rating	Highest Rated Tool(s)	Lowest Rated Tool(s)
Memory	3.9/5	Google Calendar, Phone Reminders, Todoist (all 5/5)	ChatGPT (2/5), Paper Planners (2/5)
Action	3.7/5	Pomodoro Apps (5/5), ChatGPT (4/5)	Notion (2/5)
Usability	3.7/5	Phone Reminders, Todoist (both 5/5)	Notion (2/5)
Focus	3.4/5	Pomodoro Apps, Paper Planners (both 5/5)	Notion (2/5)
Effort	3.3/5	Phone Reminders, ChatGPT (both 5/5)	Notion (2/5)
Activation	2.9/5	Phone Reminders, ChatGPT (both 5/5)	Notion (2/5)
Emotion	2.9/5	ChatGPT, Paper Planners, Habit Trackers (all 5/5)	Notion (2/5)

Key Finding: Tools systematically excel at Memory support but fail at Activation and Emotion—the components most critical for students with executive dysfunction.

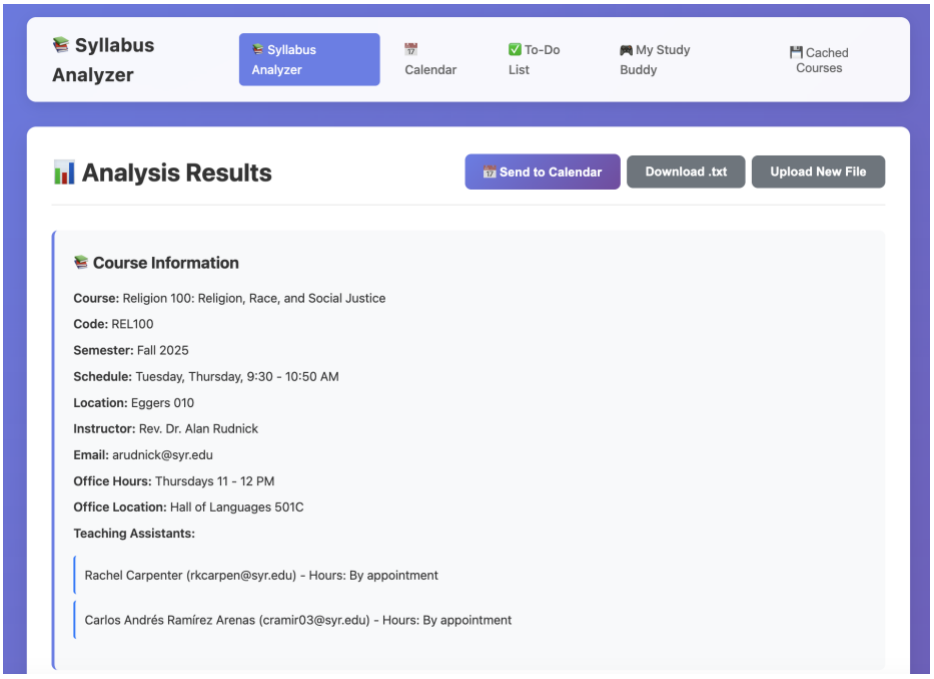
Appendix D: Website Interface Screenshots

Figure D.1: Syllabus Processor - Upload Interface



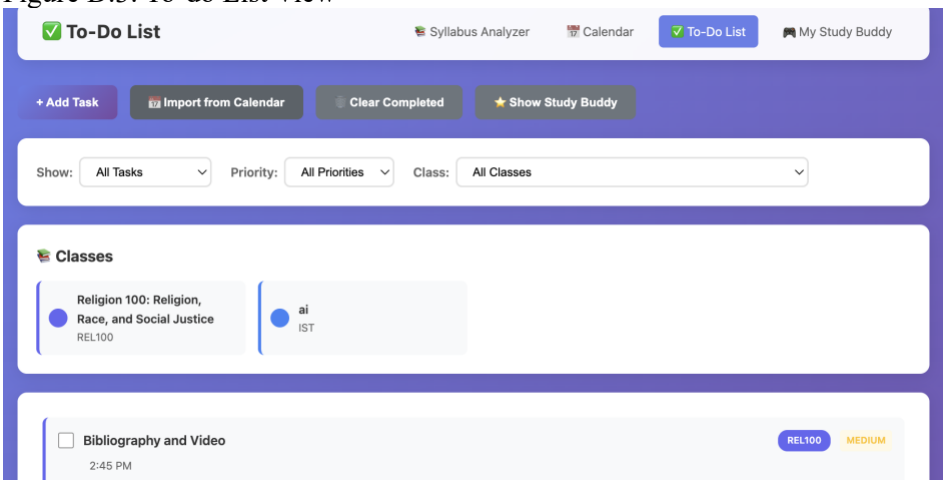
Syllabus Processor upload interface featuring drag-and-drop functionality

Figure D.2: Syllabus Processor - Review Screen



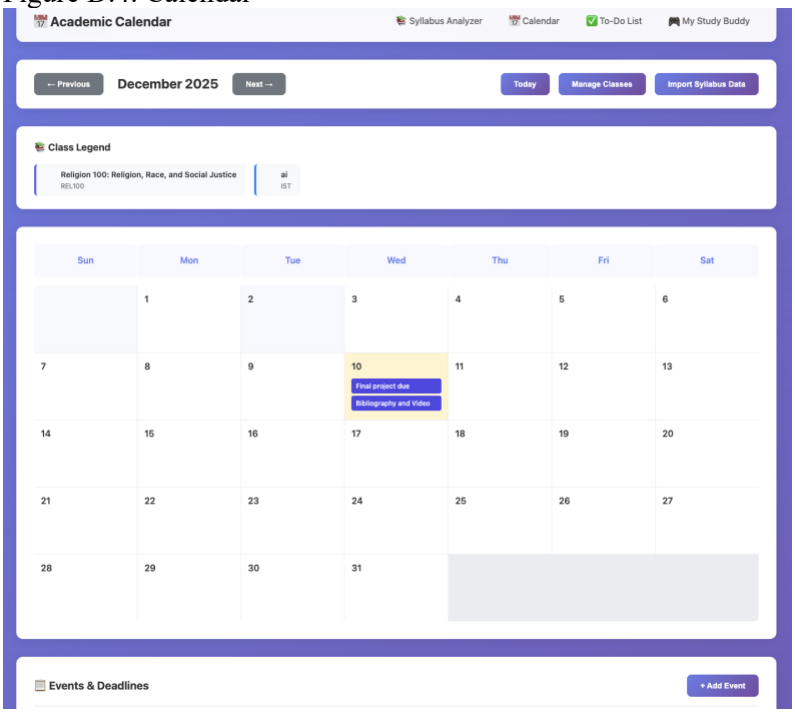
Assignment review screen displaying extracted course information, due dates, and assignment descriptions before final import.

Figure D.3: To-do List View



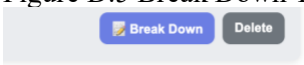
Priority Task List showing algorithm-generated top three tasks with energy level indicators, time estimates, and one-click "Start Now" buttons.

Figure D.4: Calendar



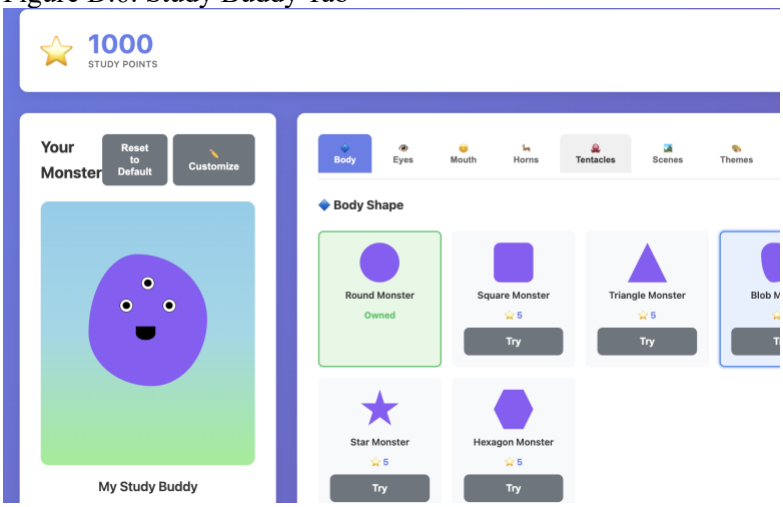
Calendar interface demonstrating one of the views (Month).

Figure D.5 Break Down Tool



Displays how to access the break down tool.

Figure D.6: Study Buddy Tab



Study Buddy feature interface showing non-invasive point system, customizable buddy character, and theme selection options.